

1. Administrative & Study Identification

Full Study Title: Study of Ulevostinag (MK-1454) Alone or in Combination With Pembrolizumab (MK-3475) in Participants With Advanced/Metastatic Solid Tumors or Lymphomas (MK-1454-001)
Short Title/Acronym: MK-1454-001
Protocol Number: MK-1454-001
NCT Number: NCT03010176
Posting ID: 1048756961226174809
Sponsor Name: Merck (Merck & Co., Inc. / Merck Sharp & Dohme LLC)
Investigational Product: Ulevostinag (MK-1454, STING agonist) and Pembrolizumab (MK-3475, Trade Name: Keytruda, ATC Code: L01FF02, [EMA Product Information](#))
Phase of Development: Phase 1
Trial Status: Completed
Medical Monitor: Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: In Part 1, to identify a maximum tolerated dose (MTD) or maximum administered dose (MAD) of ulevostinag alone and of ulevostinag in combination with pembrolizumab in participants with advanced/metastatic solid tumors or lymphomas. In Part 2, to evaluate the safety and efficacy of ulevostinag via intratumoral (IT) injection in combination with pembrolizumab in selected solid tumors at the Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To characterize the pharmacokinetics (PK) of ulevostinag, specifically Area Under the Curve (\$AUC0-24\$), minimum plasma concentration (\$Cmin\$), and maximum plasma concentration (\$Cmax\$).

2.2 Background & Rationale To address the critical need for improved therapies in treatment-refractory solid tumors and lymphomas. Ulevostinag is a cyclic dinucleotide stimulator of interferon genes (STING) agonist designed to trigger a robust immune-mediated response in the tumor microenvironment. This study evaluates whether intratumoral STING activation can safely enhance the efficacy of systemic anti-PD-1 therapy (pembrolizumab).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm/Uncontrolled (Dose-escalation and Expansion cohorts)

Blinding: Open-label **Randomization:** N/A (Sequential dose-escalation followed by expansion cohorts)

3.2 Planned Enrollment & Duration Target **N**: 156 evaluable participants **Number of Sites:** Global Multicenter **Trial Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria All Arms and Cohorts (Parts 1 and 2): 1. Age ≥ 18 years with an Eastern Cooperative Oncology Group (ECOG) Performance Status score of 0 or 1. 2. Has ≥ 1 injectable lesion which is measurable and amenable to injection and biopsy. 3. Demonstrates adequate organ function within 7 days prior to treatment initiation. 4. Highly effective contraception or abstinence required during intervention and for at least 130 days after the last dose. 5. HIV-infected participants must have well-controlled HIV on ART (CD4+ T-cell count >350 cells/mm³, virologic suppression, and stable regimen for ≥ 4 weeks prior to Day 1). **All Part 1 Arms:** 6. Has ≥ 1 distant, discrete non-injected lesion which is amenable to biopsy and measurable (RECIST 1.1 or revised IWG criteria). **Part 1 Arm 1 & Arm 2 (Cutaneous/Subcutaneous Lesions):** 7. Histologically- or cytologically-confirmed advanced/metastatic solid tumor or lymphoma, having received/been intolerant to all treatments known to confer clinical benefit. Stage III or IV disease that is not surgically resectable. Stage IIB (T3N0M0B0-1) cutaneous T cell lymphoma (CTCL) participants are eligible. **Part 1 Arm 3 (Visceral Lesions):** 8. Stage III or IV disease not surgically resectable. Metastatic liver involvement must not exceed one third of the total liver volume. Hepatocellular carcinoma is excluded. **Part 2 Expansion Cohorts:** 9. **Cohort A (HNSCC Anti-PD-1/PD-L1 Refractory):** HNSCC of the oral cavity, oropharynx, hypopharynx, or larynx (excluding nasopharynx). Stage III, IVa, or IVb with recurrent/persistent disease after chemoradiation, refractory to platinum-based chemotherapy and anti-PD-1/PD-L1 therapy; OR Stage IVc refractory to the same. 10. **Cohort B (Anti-PD-1/PD-L1 TrT-Naïve or Refractory TNBC):** Unresectable locally advanced or metastatic TNBC. Received ≥ 1 prior systemic treatment, treated with anthracycline and/or taxane, and has LDH $< 2.5 \times$ upper limit of normal (ULN). 11. **Cohort C (Anti-PD-1/PD-L1 TrT-Naïve Solid Tumors-Liver):** Stage IV solid tumor not surgically resectable. Resolution of prior chemotherapy toxicities to Grade 1 or baseline.

4.2 Exclusion Criteria 1. Received chemotherapy, definitive radiation, or biological cancer therapy within 4 weeks prior to first dose. 2. Received an investigational agent/device within 28 days of ulevostinag administration. 3. Requires any other form of antineoplastic therapy while on study. 4. On chronic systemic steroid therapy in excess of replacement doses (prednisone > 10 mg/day) or any immunosuppressive medication. 5. History of a

second malignancy, unless potentially curative treatment completed and disease-free for 2 years. 6. Clinically active central nervous system (CNS) metastases and/or carcinomatous meningitis. 7. Severe hypersensitivity reaction to a monoclonal antibody. 8. Active autoimmune disease requiring systemic treatment in the past 2 years, or history of vasculitis. 9. Active infection requiring therapy, Hepatitis B or C infection, or HIV with history of Kaposi's sarcoma, multicentric Castleman's disease, or opportunistic infection within 6 months. 10. History of non-infectious pneumonitis requiring steroids or current pneumonitis. 11. Prior allogeneic hematopoietic stem cell transplantation within 5 years. 12. Pregnant, breastfeeding, or expecting to conceive/father children. 13. Received a live vaccine within 30 days prior to first dose. 14. History of re-irradiation for HNSCC at the projected injection site. Tumor encasing a major blood vessel, or ulceration/fungation at the injection site. 15. Prior treatment with a STING agonist (e.g., ulevostinag, ADU-S100). **Part 2 Expansion Cohorts Only:** 16. Weight loss $> 10\%$ over 2 months prior to first dose. 17. Clinically relevant ascites (requiring paracentesis) or moderate radiographic ascites. 18. History of interstitial lung disease. 19. Participants with MSI-H CRC are excluded from Part 2 Cohort C.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - Part 1:

Participants allocated to Ulevostinag monotherapy (cutaneous/subcutaneous lesions), Ulevostinag + Pembrolizumab (cutaneous/subcutaneous lesions), or Ulevostinag + Pembrolizumab (visceral lesions). - **Part 2 (Cohorts A, B, and C):** Ulevostinag via IT injection at the RP2D PLUS Pembrolizumab via IV infusion. **Route of Administration:** Intratumoral (IT) injection for Ulevostinag; Intravenous (IV) infusion for Pembrolizumab. **Duration of Treatment:** Administered for up to 35 cycles (approximately 2 years). **Observation Period:** Participants undergo an inpatient observation period following the first dose of ulevostinag on Cycle 1 Day 1: at least 24 hours in Part 1, and at least 8 hours in Part 2.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care and symptom management medications. Prior exposure to immunotherapeutics (including PD-1/PD-L1 inhibitors) is allowed. **Prohibited:** Concurrent antineoplastic therapy, live vaccines within 30 days, high-dose systemic corticosteroids, and other investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent,

Tumor Imaging, Baseline Biopsy, Labs | | **Baseline** | Cycle 1 Day 1 | First IT Injection, First IV Infusion (if combo arm), Post-dose Observation (24h for Part 1, 8h for Part 2) | | **Interim** | Every 21 Days (3 Weeks) | Safety Labs, IT Injections, PK sampling, Tumor imaging | | **End of Study** | Up to 2 Years | End of Treatment assessment (max 35 cycles), Survival and progression follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Part 1:** Percentage of Participants Who Experienced a Dose-limiting Toxicity (DLT) Per Common Terminology Criteria for Adverse Events, Version 4.0 (CTCAE 4.0). Assessed during the first cycle (21 days). 2. **Parts 1 and 2:** Number of Participants Who Experienced One or More Adverse Events (AEs). 3. **Parts 1 and 2:** Number of Participants Who Discontinued Study Drug Due to an AE.

7.2 Secondary Endpoints 1. **Parts 1 and 2:** Ulevostinag Area Under the Plasma Drug Concentration-Time Curve From Time Zero to 24 Hours (\$AUC₀₋₂₄\$). 2. **Parts 1 and 2:** Ulevostinag Minimum Plasma Concentration (\$C_{min}\$). 3. **Parts 1 and 2:** Ulevostinag Maximum Plasma Concentration (\$C_{max}\$).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation throughout the trial and following the last dose of study treatment. AEs include any unfavorable and unintended sign temporally associated with the use of study treatment. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** A Safety Review Committee (SRC) oversees dose-escalation decisions.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: The All-Patients-as-Treated (APaT) / Safety Population is used for safety and PK evaluations. Efficacy is analyzed in participants who receive \$\\ge 1\$ dose and have measurable baseline disease. **Sample Size Justification:** Driven by the dose-escalation statistical framework to establish safety/MTD/MAD, followed by expansion cohorts. **Handling of Missing Data:** Efficacy analyses are based on observed data; missing PK and biomarker data are generally not imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring to ensure protocol adherence, AE reporting compliance, and proper IT injection technique. **Audit Trail:** Full audit trails maintained via validated Electronic Data Capture (EDC) systems with built-in data clarification forms.

11. Study Identifiers & Governance

Primary ID: MK-1454-001 **Registry Number:** NCT03010176 **Posting ID:** 1048756961226174809 **Sponsor/Lead Organization:** Merck (Merck & Co., Inc.) **Study Title:** Study of Ulevostinag (MK-1454) Alone or in Combination With Pembrolizumab (MK-3475) in Participants With Advanced/Metastatic Solid Tumors or Lymphomas (MK-1454-001) **Official Regulatory Title:** Phase 1 Open-label, Multicenter Study of MK-1454 Administered by Intratumoral Injection as Monotherapy and in Combination With Pembrolizumab for Patients With Advanced/Metastatic Solid Tumors or Lymphomas

12. Clinical Framework (Oncology Specific)

Primary Condition: Solid Tumors, Lymphoma **Diagnosis Threshold:** Histologically/cytologically confirmed unresectable/metastatic disease refractory to prior therapies **Medical Methodology:** Intratumoral (IT) STING Agonism + Intravenous (IV) PD-1 Checkpoint Inhibition **Optimization Target:** Reversal of tumor immune evasion and identification of MTD/RP2D

13. Study Procedures & Methodology

Management Pathway: Targeted Intratumoral microenvironment immunomodulation **Education Protocol (HCP):** Specialized training for interventional radiologists/oncologists on IT injection guidance for cutaneous, subcutaneous, and visceral (e.g., liver) lesions **Education Protocol (Patient):** Counseling regarding immune-mediated adverse events (imAEs), inpatient observation expectations, and injection-site reaction management **Quality Audit Mechanism:** Centralized independent review of radiological imaging to confirm RECIST 1.1 / IWG response criteria

14. Observation & Follow-Up Schedule

Total Duration: Up to 35 cycles (approximately 2 years) of active treatment plus post-treatment survival follow-up **Onsite Visit Cadence:** Every 21 days matching the treatment cycles **Remote Monitoring Frequency:** Routine safety follow-up calls between cycles as clinically indicated **Checklist Review Interval:** Tumor imaging assessment performed systematically per protocol timelines

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					